

A finite cover for coefficient positivity of stretched Littlewood–Richardson polynomials in the seven-row, size-thirty box

Abstract

We give a finite cover argument for nonnegativity of the ordinary monomial coefficients of every stretched Littlewood–Richardson polynomial with partition lengths at most seven and outer size at most thirty. Explicit reductions leave 358,952 residual triples. The computational part consists of exact finite enumeration, local reduction checks and rational Ehrhart polynomial computations. Its three named dependencies are stated precisely below, separately from the mathematical implication they establish.

1 Statement and mathematical inputs

Partitions are finite weakly decreasing sequences of positive integers, with zero padding allowed. A triple is balanced if $|\lambda| = |\mu| + |\nu|$; the outer partition is λ . The box in the Epoch problem [1] consists of balanced triples with all lengths at most seven and outer size at most thirty. Balance implies the other size bounds. **Standing convention.** Every triple in a polynomial identity below is a balanced triple of partitions; every displayed part beyond a partition’s length is zero. Subtracting parts is permitted only when the displayed hypotheses give nonnegative weakly decreasing sequences, after which trailing zeros are removed. A rank is an integer at least the three lengths.

Write $P_T(t) = P_{\mu\nu}^\lambda(t)$ for the unique polynomial over $\mathbb{Q}$ satisfying

$$\forall t \in \mathbb{Z}_{\geq 1}, \quad P_T(t) = c_{t\mu, t\nu}^{t\lambda}.$$

For every balanced $T = (\lambda, \mu, \nu)$ with $\max(\ell(\lambda), \ell(\mu), \ell(\nu)) \leq 7$ and $|\lambda| \leq 30$, our target is

$$\forall i \in \mathbb{Z}_{\geq 0}, \quad [t^i]P_T(t) \geq 0.$$

Here $[t^i]$ always means the coefficient in $1, t, t^2, \dots$; coefficients past the degree are zero. The theorem quantifies over every positive stretch, including stretches whose resulting partitions exceed the box. The zero polynomial is characterized by its positive-integer values; when $c_{\mu\nu}^\lambda = 0$, its value at zero is 0, even though the all-empty triple has LR coefficient 1.

We use the Littlewood–Richardson tableau and representation rules, and exchange of the inner partitions. Polynomiality and the bound $\deg P_T \leq \binom{n-1}{2}$ for rank $n \geq 2$ follow from Rassart [2, Corollary 4.2], following Derksen–Weyman. For ranks zero and one the polynomial is constant. The following established implications hold for all balanced triples:

$$\frac{c_{\mu\nu}^\lambda}{P_T(t)} \left| \begin{array}{ccc} 0 & 1 & 2 \\ 0 & 1 & t+1. \end{array} \right.$$

The first is the contrapositive of saturation [3]; the second is Fulton's theorem [4, Section 6.1]; the third is Ikenmeyer's theorem [5, Theorem 1.1]. For positive base multiplicity, $P_T(0) = 1$ by the Ehrhart interpretation. No positivity conjecture or earlier computational sweep is an input. The tensor argument below also uses the standard complete reducibility of finite-dimensional rational $GL_n(\mathbb{C})$ representations, duality of tensor products and the tensor–Hom adjunction; see Milne [7, 17.14 and 20.35] for the complete reducibility input.

For $0 \leq B \leq 30$, let $\mathcal{B}_B$ mean nonnegativity for all balanced triples in the box with outer size at most B . Only $\mathcal{B}_0$ is needed: the all-empty triple has polynomial 1. Agreement of polynomials at all positive integers implies equality, since a nonzero polynomial has at most its degree many roots.

Computational convention. This is a conventional computer-assisted proof: its finite assertions are established by the complete exact computation accompanying the article. The exact arithmetic implementations and their execution are trusted, as specified below. The mathematical identities and finite coverage argument are proved here; no completed proof-assistant verification is asserted.

2 Tableau arrays and shortening

Lemma 1 (Array description). *Suppose $\mu \subseteq \lambda$ and all partitions have length at most n . At stretch $t \geq 1$, LR tableaux correspond to nonnegative integer arrays $x_{j,i}$, $1 \leq j, i \leq n$, satisfying*

$$\sum_i x_{j,i} = t(\lambda_j - \mu_j), \quad \sum_j x_{j,i} = t\nu_i, \quad (1)$$

$$x_{j,i} = 0 \quad (i > j), \quad (2)$$

$$\sum_{h < j} x_{h,i} - \sum_{h \leq j} x_{h,i+1} \geq 0 \quad (1 \leq i < n), \quad (3)$$

$$t(\mu_{j-1} - \mu_j) + \sum_{i < b} x_{j-1,i} - \sum_{i \leq b} x_{j,i} \geq 0 \quad (2 \leq j \leq n, \ 1 \leq b \leq n). \quad (4)$$

Define

$$q_k = \sum_{j \leq k} (\mu_j + \nu_j - \lambda_j).$$

The number of labels at most k below the first k rows is tq_k . Existence of a tableau therefore implies $q_k \geq 0$. It also implies

$$\lambda_j - \mu_j \leq \nu_1. \quad (5)$$

Proof. The array records each row's label counts; weak increase determines the filling uniquely.

For labels $i, i+1$, their smallest prefix-count difference within row j occurs after its $i+1$'s and before its i 's are read. This is the left side of (3). These inequalities are therefore equivalent to the lattice condition.

The first row contains only ones. Inductively, if earlier rows contain only labels at most $j-1$, an entry $i > j$ in row j would be read before any $i-1$ in that row, and earlier rows contain no $i-1$. This violates the lattice condition and proves (2).

The entries at most b in row j end at column

$$t\mu_j + \sum_{i \leq b} x_{j,i}.$$

The entries less than b in row $j - 1$, together with its removed inner cells, end at

$$t\mu_{j-1} + \sum_{i < b} x_{j-1,i}.$$

Column strictness is equivalent to the first endpoint not exceeding the second for each b . This gives (4).

All cells in the first k rows have labels at most k . Subtracting their number from the total content of such labels gives tq_k .

For $n > 1$, summing (3) over i gives

$$\sum_{i=2}^n x_{j,i} \leq \sum_{h < j} x_{h,1} - \sum_{h < j} x_{h,n}.$$

Adding $x_{j,1}$ bounds the row length by the total number $t\nu_1$ of ones. For $n = 1$, the content equation gives the same bound. $\square$

Theorem 2 (Strengthened shortening). *Let $\mu \subseteq \lambda$, with all partitions of length at most n . Fix $1 \leq k < n$, suppose $q_k \geq 0$, and put*

$$r_{k+1} = \lambda_{k+1} - \mu_{k+1}, \quad b_k = \min(q_k, r_{k+1}).$$

If an integer $a \geq 0$ satisfies

$$a \leq \lambda_k - \lambda_{k+1}, \quad a \leq \mu_k - \mu_{k+1} - b_k, \tag{6}$$

define

$$\lambda_j^- = \lambda_j - a \mathbf{1}_{\{j \leq k\}}, \quad \mu_j^- = \mu_j - a \mathbf{1}_{\{j \leq k\}}.$$

These are partitions after zero parts are dropped, and

$$P_{\mu\nu}^\lambda = P_{\mu^-, \nu}^{\lambda^-}. \tag{7}$$

The analogous exchanged-inner identity holds.

Proof. The bounds ensure nonnegative weakly decreasing modified sequences. Their row-length differences are unchanged.

At stretch t , the margins, support, and lattice inequalities remain identical. Only column inequalities crossing rows $k, k + 1$ change.

For $b \leq k$, the lower partial row sum is bounded by both tq_k and tr_{k+1} . The modified slack is therefore at least

$$t(\mu_k - \mu_{k+1} - a - b_k) \geq 0.$$

For $b \geq k + 1$, triangular support makes both partial sums complete row sums. The modified slack is

$$t(\lambda_k - \lambda_{k+1} - a) \geq 0.$$

Thus every original array is modified-feasible. Conversely, each original crossing slack equals the modified slack plus ta ; all other conditions coincide. This proves a bijection at every stretch. Positive base multiplicity was not assumed. $\square$

3 Deletion and factorization

Lemma 3 (Empty-row deletion). *If $\mu \subseteq \lambda$ and $\lambda_j = \mu_j$, deleting part j from both gives*

$$P_{\mu\nu}^\lambda = P_{\hat{\mu},\nu}^{\hat{\lambda}}. \quad (8)$$

If $j \leq \ell(\lambda)$, the outer size decreases by $\lambda_j > 0$.

Proof. Put $L = \lambda_j = \mu_j$. At stretch t , row j is empty. All skew cells above it lie in columns greater than tL , because their inner row lengths are at least tL . All skew cells below it lie in columns at most tL , because their outer row lengths are at most tL .

Deleting the empty row introduces no column comparisons across the cut, since the two groups occupy disjoint column ranges. Other comparisons and the reading word remain unchanged. Inserting the row reverses the construction. $\square$

Theorem 4 (Initial-sum factorization). *Suppose explicitly that $|\lambda| = |\mu| + |\nu|$, $\mu \subseteq \lambda$, and $1 \leq k < n$ for a rank n , with $q_k = 0$. Splitting all three partitions after part k gives*

$$P_{\mu\nu}^\lambda = P_{\mu_{\leq k}, \nu_{\leq k}}^{\lambda_{\leq k}} P_{\mu_{> k}, \nu_{> k}}^{\lambda_{> k}}. \quad (9)$$

Both triples are balanced. If $1 \leq k < \ell(\lambda)$, both outer sizes are strictly smaller.

Proof. The first k rows use only labels at most k . Since $q_k = 0$, they exhaust those labels. The lower rows use larger labels. Restriction to the upper rows and subtraction of k from the lower labels produces tableaux of the displayed triples.

Conversely, concatenate a pair of such tableaux, increasing the lower labels by k . Column strictness across the cut follows from the separation of the label ranges. Lattice comparisons within each range are inherited. For the remaining comparison, the upper rows supply $t\nu_k$ copies of k , while every lower prefix has at most $t\nu_{k+1}$ copies of $k+1$. The inequality follows from $\nu_k \geq \nu_{k+1}$.

This is a bijection at every stretch. Balance follows from $q_k = 0$ and total balance. A proper cut within the positive outer parts gives two positive, smaller outer sizes.

If both factors have nonnegative monomial coefficients, so does their product, since each product coefficient is a sum of products of nonnegative numbers. Hence a negative coefficient in the product requires a negative coefficient in a factor. $\square$

Lemma 5 (Full columns and scaling). *If $\ell(\lambda), \ell(\mu), \ell(\nu) \leq n$, $\mu \subseteq \lambda$, and $\lambda_n, \mu_n \geq 1$, subtracting 1 from all n parts of λ and μ preserves P . The exchanged-inner version holds as well. If all parts are divisible by $g \geq 1$, and Q is the divided triple's polynomial, then*

$$P(t) = Q(gt), \quad [t^i]P = g^i[t^i]Q. \quad (10)$$

Proof. Full-column removal translates every skew cell one column left. At stretch t , translate by t columns. The reading word and relative column comparisons are unchanged.

For division, the original triple stretched by t is the divided triple stretched by gt . This proves the polynomial identity. Each g^i is positive, so coefficient signs are preserved. $\square$

4 Six representatives

Lemma 6 (Rectangular complement). *For $\ell(\alpha) \leq n$ and $M \geq \alpha_1$, put*

$$\alpha^{c,M} = (M - \alpha_n, \dots, M - \alpha_1).$$

Then

$$s_{\alpha^c, M}(x_1, \dots, x_n) = (x_1 \cdots x_n)^M s_{\alpha}(x_1^{-1}, \dots, x_n^{-1}), \quad (11)$$

and consequently

$$V_{\alpha}^* \cong \det^{-M} \otimes V_{\alpha^c, M}. \quad (12)$$

Proof. Regard a tableau as M columns, allowing empty columns. Complement each column's entry set in $\{1, \dots, n\}$ and reverse the column order.

For adjacent original columns with sets A, B , row weak increase is equivalent to

$$|A \cap \{1, \dots, r\}| \geq |B \cap \{1, \dots, r\}| \quad (1 \leq r \leq n).$$

Complementation reverses these inequalities, so the reversed complemented columns satisfy row weak increase. The shape becomes $\alpha^{c, M}$, and the operation is an involution.

If label i occurred e_i times, it now occurs $M - e_i$ times. Summing weights proves (11). Taking dual characters and using the supplied irreducible Schur-character interpretation gives (12). $\square$

Theorem 7 (Six polynomial-preserving representatives). *Suppose $c_{\mu\nu}^{\lambda} > 0$, pad to n , and assume $\mu_n = \nu_n = 0$. Write*

$$N = |\lambda|, \quad m = |\mu|, \quad v = |\nu|, \quad L = \lambda_1, \quad h = \lambda_n, \quad u = \mu_1, \quad w = \nu_1.$$

Six balanced partition triples of length at most n have polynomial $P_{\mu\nu}^{\lambda}$ and outer sizes

$$N, \quad nL - m, \quad nL - v, \quad n(u + w) - N, \quad n(u - h) + v, \quad n(w - h) + m. \quad (13)$$

Proof. For $K \geq W_1$, define

$$O_K(W) = (K - W_n, \dots, K - W_1).$$

Set $A = \mu$, $B = \nu$, $C = \lambda^{c, L}$. Their last parts are zero and their sizes sum to nL .

For $G = GL_n$, the representation interpretation and tensor-Hom identification give

$$c_{\mu\nu}^{\lambda} = \dim \operatorname{Hom}_G^L(\det, V_A \otimes V_B \otimes V_C). \quad (14)$$

Multiplicity equals the indicated Hom dimension because equivariant maps between distinct irreducibles vanish and an equivariant endomorphism of an irreducible is scalar. For the latter statement, an eigenspace of an equivariant endomorphism is a nonzero invariant subspace and hence the whole irreducible.

Expression (14) is symmetric in A, B, C . Their first parts are at most L : containment gives this for A, B , and $C_1 = L - h$. Choosing one as W , outer partition $O_L(W)$, and the other two as inner partitions gives the first three representatives.

For $W_n = 0$, put

$$W^* = (W_1 - W_n, \dots, W_1 - W_1).$$

Dualizing each irreducible summand preserves the corresponding dual multiplicity. Formula (12) shows that (14) also equals

$$\dim \operatorname{Hom}_G^{K^*}(\det, V_{A^*} \otimes V_{B^*} \otimes V_{C^*}), \quad K^* = u + w - h. \quad (15)$$

Lemma 1 gives $L \leq u + w$, $h \leq w$, and, by exchange symmetry, $h \leq u$. Thus

$$K^* - A_1 = w - h \geq 0, \quad K^* - B_1 = u - h \geq 0, \quad K^* - C_1 = u + w - L \geq 0.$$

The three further outer partitions $O_{K^*}(W)$, with W chosen from A^*, B^*, C^* , are therefore valid.

All constructions commute with stretching. Thus their polynomial identities follow from the multiplicity identities.

Finally,

$$|C| = nL - N, \quad |A^*| = nu - m, \quad |B^*| = nw - v, \quad |C^*| = N - nh.$$

Taking the pair sums in the two triples gives (13). $\square$

5 The minimum-size domain

Theorem 8 (Necessary conditions). *Assume $\mathcal{B}_B$, and choose a counterexample of minimum outer size N , if one exists. Put $n = \ell(\lambda)$. Then:*

1. $B < N \leq 30$, the base multiplicity is positive, and the gcd of all positive parts is one.
2. $\mu_n = \nu_n = 0$, and $q_k > 0$ for $1 \leq k < n$.
3. $\lambda_j > \mu_j, \nu_j$ for every $j \leq n$.
4. $\mu_k, \nu_k \leq \lambda_{k+1}$ for $k < n$.
5. For every $1 \leq k < n$ with $\lambda_k > \lambda_{k+1}$,

$$\begin{aligned} \mu_k - \mu_{k+1} &\leq \min(q_k, \lambda_{k+1} - \mu_{k+1}), \\ \nu_k - \nu_{k+1} &\leq \min(q_k, \lambda_{k+1} - \nu_{k+1}). \end{aligned} \tag{16}$$

6. All six sizes in (13) are at least N , equivalently

$$\max(m, v) \leq nL - N, \quad 2N \leq n(u + w), \quad m \leq n(u - h), \quad v \leq n(w - h). \tag{17}$$

Proof. The supplied zero-multiplicity implication gives positive base multiplicity. The hypothesis $\mathcal{B}_B$ gives $N > B$.

Common division or an applicable full-column removal would preserve a negative coefficient and decrease the outer size. This proves the gcd assertion and the zero last parts.

The deficits are nonnegative by Lemma 1. A zero proper deficit permits factorization by Theorem 4; some smaller factor would have a negative coefficient. Hence the proper deficits are positive.

Equality $\lambda_j = \mu_j$ permits deletion of a positive outer part by Lemma 3; equality with ν_j is treated by exchange symmetry. Thus both containments are strict.

Failure of (16) at a positive outer gap allows $a = 1$ in Theorem 2, giving a smaller counterexample. At such a gap, (16) implies

$$\mu_k - \mu_{k+1} \leq \lambda_{k+1} - \mu_{k+1},$$

hence $\mu_k \leq \lambda_{k+1}$. At a zero outer gap, containment gives the same conclusion. Exchange symmetry handles ν .

Finally, a symmetry representative of outer size less than N would automatically remain in the box: its length is at most n , and each inner size is bounded by its balanced outer size. Minimality therefore makes all six sizes at least N . Rearranging these inequalities gives (17). $\square$

Corollary 9 (The containing partition and deficit). *Under Theorem 8, put $\lambda_{n+1} = 0$ and*

$$\kappa_j = \min(\lambda_j - 1, \lambda_{j+1}), \quad e = \#\{j < n : \lambda_j = \lambda_{j+1}\}.$$

Then κ is a partition with last part zero,

$$\mu, \nu \subseteq \kappa, \quad |\kappa| = N - L - e, \quad (18)$$

and

$$s = N - 2(L + e) \geq 0, \quad (|\kappa| - |\mu|) + (|\kappa| - |\nu|) = s. \quad (19)$$

If $s = 0$, necessarily $\mu = \nu = \kappa$. If $s = 1$, up to exchange, one inner partition is κ and the other is obtained by removing one removable corner.

Proof. The coordinatewise minimum of the nonnegative decreasing sequences $\lambda_j - 1$ and λ_{j+1} is nonnegative and decreasing. Its last part is zero.

Strict containment gives $\mu_j, \nu_j \leq \lambda_j - 1$; interlacing gives the other bound needed for containment in κ . At a positive outer gap, $\kappa_j = \lambda_{j+1}$. At an equality, $\kappa_j = \lambda_{j+1} - 1$. Summing proves (18).

Subtracting $|\mu| + |\nu| = N$ from $2|\kappa|$ gives (19). Both deficits are nonnegative. Deficit zero forces equality of both subpartitions with κ ; deficit one gives the stated removable-corner description. $\square$

6 Array notation and bounds

Use Lemma 1 at stretch one and write

$$S_{j,i} = \sum_{h \leq j} x_{h,i}, \quad S_{0,i} = 0, \quad y_{j,b} = \mu_j + \sum_{i \leq b} x_{j,i}, \quad y_{j,0} = \mu_j.$$

The lattice and column slacks are respectively

$$L_{j,i} = S_{j-1,i} - S_{j,i+1} \geq 0 \quad (1 \leq j \leq n, 1 \leq i < n), \quad (5)$$

$$C_{j,b} = y_{j-1,b-1} - y_{j,b} \geq 0 \quad (2 \leq j \leq n, 1 \leq b \leq n). \quad (6)$$

Lemma 10. *For an array in Lemma 1, put*

$$R_j(b) = \sum_{i=b}^n x_{j,i}.$$

Then

$$R_j(b) \leq S_{j,b} \leq \nu_b. \quad (7)$$

Whenever $j > b$,

$$y_{j,b} \leq \mu_{j-b}. \quad (8)$$

Consequently, for $p, q \geq 1$,

$$\lambda_{p+q+1} \leq \mu_{p+1} + \nu_{q+1}, \quad (9)$$

where a part beyond the chosen length is zero.

Proof. For $b < n$, the lattice inequalities give

$$x_{j,i+1} \leq S_{j-1,i} - S_{j-1,i+1}.$$

Summing over $i = b, \dots, n-1$ yields

$$\sum_{i=b+1}^n x_{j,i} \leq S_{j-1,b} - S_{j-1,n} \leq S_{j-1,b}.$$

Adding $x_{j,b}$ proves the first inequality in (7). For $b = n$, it is simply $x_{j,n} \leq S_{j,n}$. The content condition proves the second inequality.

Iterating the column inequalities b times gives

$$y_{j,b} \leq y_{j-1,b-1} \leq \dots \leq y_{j-b,0} = \mu_{j-b},$$

proving (8).

If $p+q+1 \leq n$, the two established bounds give

$$\lambda_{p+q+1} = y_{p+q+1,q} + R_{p+q+1}(q+1) \leq \mu_{p+1} + \nu_{q+1}.$$

Otherwise the left side is zero. □

7 Rectangular shortening

For $k \geq 1$, write $\alpha - a(1^k)$ for subtraction of a from the first k parts of α .

Theorem 11. *Suppose λ, μ, ν are balanced partitions with $\mu, \nu \subseteq \lambda$. Let $p, q \geq 1$, put $r = p+q$, and let $a \geq 1$ be an integer satisfying*

$$\mu_p - a \geq \mu_{p+1}, \quad \nu_q - a \geq \nu_{q+1}, \quad \lambda_r - a \geq \max\{\lambda_{r+1}, \mu_{p+1} + \nu_{q+1}\}. \quad (10)$$

Define

$$\widehat{\lambda} = \lambda - a(1^r), \quad \widehat{\mu} = \mu - a(1^p), \quad \widehat{\nu} = \nu - a(1^q).$$

These are balanced partitions, and

$$P_{\mu\nu}^\lambda = P_{\widehat{\mu}\widehat{\nu}}^{\widehat{\lambda}}. \quad (11)$$

Proof. The inequalities ensure weak decrease and nonnegativity. The three sizes decrease by ra , pa , and qa ; hence balance is preserved.

Containment of $\widehat{\mu}$ in $\widehat{\lambda}$ is unchanged in rows at most p and beyond r . For $p < j \leq r$,

$$\lambda_j - a \geq \lambda_r - a \geq \mu_{p+1} + \nu_{q+1} \geq \mu_j.$$

The containment argument for $\widehat{\nu}$ uses $q < j \leq r$ and $\nu_j \leq \nu_{q+1}$.

Choose $n \geq r$ containing all parts, and put

$$M = \mu_{p+1}, \quad N = \nu_{q+1}.$$

Given an original LR array x , define

$$\widehat{x}_{p+i,i} = x_{p+i,i} - a \quad (1 \leq i \leq q), \quad (12)$$

leaving all other entries unchanged.

We first prove nonnegativity. By (7),

$$y_{r,q} = \lambda_r - R_r(q+1) \geq \lambda_r - N \geq M + a.$$

Every column $M+1, \dots, M+a$ contains skew boxes in all rows $p+1, \dots, r$: its index exceeds the relevant inner parts and is at most λ_r . Its bottom entry is at most q . A strictly increasing column of q positive entries whose bottom entry is at most q must be $1, 2, \dots, q$. Thus $x_{p+i,i} \geq a$ for every $1 \leq i \leq q$.

The new row sums are correct: rows $p+1, \dots, r$ lose a boxes, while the skew row lengths of rows at most p and beyond r are unchanged. Each label $1, \dots, q$ loses a occurrences, giving the new content.

We calculate the changes in every inequality. For label $i < q$, subtraction in label i occurs in row $p+i$, while subtraction in label $i+1$ occurs in row $p+i+1$. Their effects on (5) cancel because

$$p+i < j \iff p+i+1 \leq j.$$

Therefore

$$\widehat{L}_{j,i} = \begin{cases} L_{j,i} - a, & i = q, j \geq r+1, \\ L_{j,i}, & \text{otherwise.} \end{cases} \quad (13)$$

For labels $i > q$ no entry of either label changes. For column inequalities with $2 \leq j \leq p$, both inner boundary terms decrease by a and neither row prefix changes. For $j \geq r+2$, neither boundary term nor row prefix changes. For the column inequalities, at $j = p+1$ the decrease of μ_p cancels the decrease of the lower row prefix, for every $b \geq 1$. For $p+2 \leq j \leq r$, the modified upper entry has label $j-p-1$ and the modified lower entry has label $j-p$. Both enter the respective prefixes exactly when $b \geq j-p$, so their effects cancel. At $j = r+1$, only the upper entry of label q is modified. Thus

$$\widehat{C}_{j,b} = \begin{cases} C_{j,b} - a, & j = r+1, b \geq q+1, \\ C_{j,b}, & \text{otherwise.} \end{cases} \quad (14)$$

The exceptional cases in these two formulas are absent if $r = n$.

Every decreased inequality has sufficient slack. By (8),

$$y_{r,q-1} \leq M.$$

For $q = 1$, this is the identity $y_{r,0} = \mu_{p+1} = M$. Hence (7) gives

$$S_{r,q} \geq R_r(q) = \lambda_r - y_{r,q-1} \geq \lambda_r - M.$$

For $j \geq r+1$,

$$L_{j,q} = S_{j-1,q} - S_{j,q+1} \geq S_{r,q} - N \geq \lambda_r - M - N \geq a.$$

If $r < n$, fix $b \geq q+1$. Summing the lattice inequalities in row $r+1$ over $i = q+1, \dots, b-1$ and adding $x_{r+1,q+1}$ gives

$$\sum_{i=q+1}^b x_{r+1,i} \leq S_{r+1,q+1} - S_{r,b}.$$

For $b = q+1$, the sum of inequalities is empty and this relation is equality. Combining it with (7) yields

$$R_r(b) + \sum_{i=q+1}^b x_{r+1,i} \leq S_{r+1,q+1} \leq N.$$

Also $y_{r+1,q} \leq M$ by (8). Consequently

$$\begin{aligned} C_{r+1,b} &= \lambda_r - R_r(b) - y_{r+1,q} - \sum_{i=q+1}^b x_{r+1,i} \\ &\geq \lambda_r - M - N \geq a. \end{aligned}$$

This proves every required new inequality.

Conversely, start with a valid hatted array and add a at exactly the positions $(p+i, i)$, $1 \leq i \leq q$. Nonnegativity is preserved and the original margins are restored. The displayed change formulas are linear identities for any pair of arrays related by the stated additions or subtractions; their derivation did not assume feasibility. Thus equations (13) and (14), read backwards, show that every inequality is unchanged or increased by a . Thus addition gives a valid original array and is inverse to (12).

We have a bijection for the original LR coefficients. Applying it to the uniformly stretched partitions with subtraction amount ta proves equality at every integer $t \geq 1$. Polynomial uniqueness gives (11). $\square$

8 Published reduction inputs

Theorem 12 (Essential Horn factorization). *Let $n \geq 2$, $|\lambda| = |\mu| + |\nu|$, $\max(\ell(\lambda), \ell(\mu), \ell(\nu)) \leq n$, and $c_{\mu\nu}^\lambda > 0$. For $1 \leq r < n$ and r -subsets I, J, K of $[n] = \{1, \dots, n\}$, write $I = \{i_1 < \dots < i_r\}$ and $\tau(I) = (i_r - r, \dots, i_1 - 1)$, dropping zeros; define $\tau(J)$ and $\tau(K)$ in the same way. Suppose*

$$c_{\tau(I), \tau(J)}^{\tau(K)} = 1, \quad \sum_{k \in K} \lambda_k = \sum_{i \in I} \mu_i + \sum_{j \in J} \nu_j.$$

For a subset, parts are selected in increasing index order; complements are taken in $[n]$. Then

$$P_{\mu\nu}^\lambda = P_{\mu_I, \nu_J}^{\lambda_K} P_{\mu_{I^c}, \nu_{J^c}}^{\lambda_{K^c}}.$$

Both factors are balanced.

Justification of the cited input. We use King–Tollu–Toumazet [8, Theorem 1.4] in the exact subset formulation recorded by Cho–Jung–Moon [11, Definition 1.2 and Theorem 1.3]. Their ordered triple (λ, μ, ν) is our (μ, ν, λ) , and their π is our τ . Their theorem assumes positive base multiplicity, $1 \leq r < n$, the multiplicity-one subset test and the displayed boundary equality, and states both the coefficient and stretched-polynomial factorizations. These are exactly the hypotheses above; regularity and strictness of the other Horn inequalities are not required. Roth [9, Reduction Theorem (3.1.1)] supplies an independent primary proof of the general representation-theoretic reduction and identifies this type- A result with KTT in his introduction. This is supporting evidence; the operative statement here is the cited subset theorem.

Positive base multiplicity persists at every positive stretch by saturation. The same subsets and their multiplicity-one test are fixed, and the boundary equality is homogeneous. Apply the coefficient theorem to each stretch, then use polynomial uniqueness. Balance of the first factor is the boundary equality; subtract it from total balance for the second factor. $\square$

For a source triple in the box, take $n = \ell(\lambda)$. Both outer factors have positive size strictly smaller than $|\lambda|$. Both lengths are at most n , so both factors are in the box. If neither factor had a negative monomial coefficient, neither could their product.

Theorem 13 (Second reduction at all stretches). *Let λ, μ, ν be balanced partitions with $\mu, \nu \subseteq \lambda$. Let $p, q \geq 1$, set $r = p + q$, and choose an integer $n \geq \max\{\ell(\lambda), \ell(\mu), \ell(\nu), r + 1\}$. Assume*

$$\mu_p > \mu_{p+1}, \quad \nu_q > \nu_{q+1}, \quad \lambda_r > \lambda_{r+1}, \quad \mu_p + \nu_q \geq \lambda_1 + \lambda_{r+1} + 1.$$

Then $\lambda^- = \lambda - (1^r)$, $\mu^- = \mu - (1^p)$ and $\nu^- = \nu - (1^q)$ are balanced partitions, and

$$\forall t \geq 1, \quad c_{t\mu, t\nu}^{t\lambda} = c_{t\mu^-, t\nu^-}^{t\lambda^-}, \quad P_{\mu\nu}^\lambda = P_{\mu^-, \nu^-}^{\lambda^-}.$$

Here t is an integer. Padding to n changes no coefficient.

Proof. The strict gaps ensure that the subtractions give partitions; balance is preserved because $r = p + q$. The unit coefficient theorem is Cho–Jung–Moon [10, Theorem 2.6], also proved in [11, Theorem 3.1]. To match the primary rectangle statement, use rectangle height n and width $W = \lambda_1$, and take its indices $\alpha = p$, $\beta = q$, $\gamma = n - r \geq 1$. The complementary outer part is $\lambda_\gamma^c = W - \lambda_{r+1}$. Its numerical hypothesis becomes $\mu_p + \nu_q + W - \lambda_{r+1} \geq 2W + 1$, exactly our inequality. Its third strict gap is $\lambda_\gamma^c > \lambda_{\gamma+1}^c$, equivalent to the outer gap. All three partitions fit this rectangle by containment. The unit theorem therefore applies with every hypothesis explicit.

Fix $t \geq 1$ and, for $0 \leq s \leq t$, put

$$\lambda^{(s)} = t\lambda - s(1^r), \quad \mu^{(s)} = t\mu - s(1^p), \quad \nu^{(s)} = t\nu - s(1^q).$$

For $s < t$, each of the three gaps is at least $t - s \geq 1$, and

$$(t\mu_p - s) + (t\nu_q - s) - (t\lambda_1 - s) - t\lambda_{r+1} = t(\mu_p + \nu_q - \lambda_1 - \lambda_{r+1}) - s \geq 1.$$

At every step where both inner partitions are contained in $\lambda^{(s)}$, apply the unit coefficient theorem, using the current rectangle width $\lambda_1^{(s)}$. In the coefficient formulation [11, Theorem 3.1], a noncontained target has coefficient zero by the usual LR convention; thus this also covers a first transition to such a target. If containment has already failed at step s , it fails at every later step: for every row j ,

$$\lambda_j^{(s)} - \mu_j^{(s)} = t(\lambda_j - \mu_j) - s \mathbf{1}_{\{p < j \leq r\}},$$

and the analogous formula for ν has q in place of p . These differences are nonincreasing in s . Consequently both coefficients in every subsequent step are zero. Every adjacent pair therefore has equal coefficients, and the final triple is $t(\lambda^-, \mu^-, \nu^-)$. Equality at every positive integer proves the polynomial identity. $\square$

9 A single least-counterexample argument for the entire box

Write $\mathcal{D}$ for the triples satisfying all the necessary conditions above with $B = 0$, together with $c_{\mu\nu}^\lambda \geq 3$. Normalize the inner order by placing the larger pair $(|\mu|, \mu)$ first: $(|\mu|, \mu) \geq (|\nu|, \nu)$ in lexicographic order. Partition tuples are compared lexicographically after deleting trailing zeros, or equivalently after padding both to length seven with zeros. If the tuples agree, either inner ordering gives the same triple. This is the normalization used by every enumeration and path check. Split $\mathcal{D} = \mathcal{D}_{\text{small}} \sqcup \mathcal{D}_{\text{large}}$ at outer size 23. No assertion of positivity through size 23 is made as an input.

For a normalized triple T , let

$$k(T) = (|\lambda|, \ell(\lambda), T)$$

with lexicographic order. The box is finite. Consequently, if it contains a polynomial with a negative coefficient, it contains one with least key. Every reduction in this manuscript either preserves P

or replaces it by Q with $P(t) = Q(gt)$, $g \geq 1$. The latter operation preserves signs because $[t^i]P = g^i[t^i]Q$. For a factorization, at least one factor inherits a negative coefficient.

Define finite sets $\mathcal{C}_{\text{small}}$ and $\mathcal{C}_{\text{large}}$ by the following explicit certificates. These definitions do not assume that a greedy selection of moves finds every possible reduction.

1. Each record of $\mathcal{D}_{\text{small}}$ is either retained in $\mathcal{C}_{\text{small}}$ or supplied with a valid essential Horn factorization, rectangular shortening, or second reduction. Each exclusion has smaller positive outer factors or a strictly smaller outer target in the original box.
2. Each record of $\mathcal{D}_{\text{large}}$ has a finite path of the proved column, division, deletion, shortening and six-representative identities. Its final normalized triple has no larger key. If its key is smaller, this is an exclusion. The label “baseline” in some data files merely records that the final outer size is at most 23; this also gives a smaller key and invokes no positivity baseline. Otherwise the path fixes the original triple.
3. Each such large fixed point is either supplied with an essential Horn factorization into smaller in-box factors, or is carried to a second path using rectangular shortening and the preceding identities. Again a smaller final key is an exclusion and an equal key is a fixed point.
4. Finally, a fixed point of the second path is either shortened by the second reduction or retained in $\mathcal{C}_{\text{large}}$.

For every path, the certificate checks its source, every local premise, every exact target, its accumulated positive scale, and its final key. Intermediate presentations may be larger than the original box; the identities have no size restriction and the final exclusion is required to lie in the original box. All intermediate partition lengths are at most seven.

Theorem 14 (Finite cover implication). *Suppose the two enumerations are complete, every certificate just specified is valid, and every polynomial attached to $\mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{large}}$ has nonnegative monomial coefficients. Then every stretched Littlewood–Richardson polynomial in the frozen box has nonnegative monomial coefficients.*

Proof. Choose a least-key negative witness T , if one exists. The empty triple has polynomial 1. The supplied theorems for base multiplicity 0, 1, and 2 give polynomials 0, 1, and $t+1$, respectively. Thus $|\lambda| \geq 1$ and the base multiplicity is at least three. The necessary-domain theorem at $B = 0$ applies to this minimum-size witness. Exchange symmetry gives its normalized representative, so $T \in \mathcal{D}_{\text{small}} \cup \mathcal{D}_{\text{large}}$.

In the small case an exclusion would produce a negative witness of strictly smaller outer size, contradicting minimality. Hence $T \in \mathcal{C}_{\text{small}}$. In the large case a nontrivial path ending at a smaller key would produce a negative witness of smaller key. A Horn split would produce a negative witness of smaller outer size. Thus neither kind of exclusion is possible. The witness must pass through both fixed-point stages and survive the final second-reduction filter. Therefore $T \in \mathcal{C}_{\text{large}}$. In either case its polynomial is nonnegative by the stated finite sign hypothesis, contradicting the choice of T . $\square$

This is a finite-box assertion. It does not prove positivity for larger partitions or for the unrestricted Littlewood–Richardson conjecture. The computational hypothesis is to be discharged by the complete, source-bound enumeration, reduction and polynomial certificates, not by observed absence in a sample, a progress counter, or a comparison of only finitely many values without a degree bound.

10 The exact finite computation

The three finite dependencies are named E (enumeration), R (reduction cover), and P (residual polynomials). They concern mathematical sets and exact rational numbers. Checksums bind their data to a particular submission; a checksum or a progress flag is not a mathematical premise.

10.1 E: complete enumeration

Here is a finite enumeration independent of all reduction choices. Let $\text{Part}(N, h, M)$ consist of weakly decreasing positive tuples of sum N , length at most h , and first part at most M . The following recursion defines it without an oracle:

$$\begin{aligned} \text{Part}(0, h, M) &= \{()\}, \\ \text{Part}(N, 0, M) &= \emptyset \quad (N > 0), \\ \text{Part}(N, h, M) &= \bigcup_{a=1}^{\min(N, M)} \{(a, \rho) : \rho \in \text{Part}(N - a, h - 1, a)\} \quad (N, h > 0). \end{aligned}$$

The entries of (a, ρ) are a followed by those of ρ . Induction on h proves that this produces precisely the stated tuples, each once: a nonempty partition determines its first part and tail uniquely. The recursion decreases h at each call.

For $N = 1, \dots, 30$, enumerate $\lambda \in \text{Part}(N, 7, 30)$; for $m = 0, \dots, N$, enumerate

$$\mu \in \text{Part}(m, 7, 30), \quad \nu \in \text{Part}(N - m, 7, 30).$$

Retain exactly the normalized triples satisfying the six items of Theorem 8 with $B = 0$ and $c_{\mu\nu}^\lambda \geq 3$. The base LR count can be computed by enumerating the nonnegative arrays of Lemma 1, with each entry bounded by $\lambda_j - \mu_j$; reject negative row bounds and count arrays satisfying all equations and inequalities. Thus the filter itself has an exact terminating definition. The implementation uses `lrcalc` and an independent tableau dynamic program to accelerate these finite counts.

The actual enumerators use two equivalent accelerations: generate inner partitions inside κ by its coordinate ceilings, or enumerate all partitions by integer-part dynamic programming and apply the literal inequalities. Corollary 9 justifies the first; the recursion above justifies the universe of the second. No search timeout or heuristic cutoff removes a tuple from E.

Proposition 15 (Finite dependency E). *The supplied small and large domain lists have no duplicate normalized triple and equal $\mathcal{D}_{\text{small}}$ and $\mathcal{D}_{\text{large}}$, respectively. Their cardinalities are 37,530 and 1,255,228.*

The independent enumeration checks equality of sets of triples and base counts, rather than cardinality alone. The small domain has outer sizes 1–23, and the large domain has outer sizes 24–30. These are counts of normalized necessary-domain triples, not counts of all ordered balanced triples in the original box.

10.2 R: local certificates and complete coverage

Every identity edge has an explicit source triple, rule name, integer parameters, target triple and positive integer scale. An accepted edge must satisfy all hypotheses of the named theorem and its exact target formula after normalization. Division has its stated scale g ; the other edges have scale one. Unknown rules, invalid parameters, malformed partitions or imbalance are rejected. Edge

sources and targets along a path must agree successively, and scales are multiplied. The resulting identity is $P_{T_0}(t) = P_{T_m}(gt)$, where g is that product. The endpoint must have key at most the source key and must be in the box; an equal key is an identical normalized triple. This also handles nontrivial loops: $P_T(t) = P_T(gt)$ does not authorize an exclusion.

A Horn record specifies its source and (r, I, J, K) . The checker validates the subset sizes and ranges, evaluates $c_{\tau(I), \tau(J)}^{\tau(K)} = 1$ by the independent finite array count, checks the boundary equality and both exact factors, and checks that their outer sizes are positive and smaller. In particular, an absent Horn entry cannot cause a false exclusion. Completeness of a catalog of all Horn inequalities is unnecessary; every used entry must pass. The rank-indexed lists in `horn_essential_catalog.json` have 3, 12, 41, 142, 521, 2042 entries for ranks 2, 3, 4, 5, 6, 7, respectively. These are direct counts of the catalog produced by `horn_catalog` in `probe_horn_primitivity.py`: it enumerates every proper subset size and every ordered triple (I, J, K) of that size, retaining exactly those with $c_{\tau(I), \tau(J)}^{\tau(K)} = 1$. The independent array-count reconstruction in `audit_core_certificates.py` checks equality of the resulting sets. The counts are reproducibility data from these finite algorithms, not an appeal to a literature table or a catalog-completeness premise.

Proposition 16 (Finite dependency R). *Every source index in E occurs exactly once in the appropriate coverage stage; the stages define the residual sets as in Theorem 14. All local rule, path endpoint and factorization checks pass. The counts are the following.*

<i>Small domain stage</i>	<i>excluded</i>	<i>remaining</i>
<i>Initial domain</i>		37, 530
<i>Horn factorization</i>	17, 883	19, 647
<i>Rectangular shortening</i>	6, 300	13, 347
<i>Second reduction</i>	932	12, 415
<i>Large domain stage</i>	<i>excluded</i>	<i>remaining</i>
<i>Initial domain</i>		1, 255, 228
<i>First paths with smaller endpoint key</i>	494, 925	760, 303
<i>Horn factorization</i>	200, 638	559, 665
<i>Second paths with smaller endpoint key</i>	196, 934	362, 731
<i>Second reduction</i>	16, 194	346, 537

In particular $|\mathcal{C}_{\text{small}}| = 12, 415$, $|\mathcal{C}_{\text{large}}| = 346, 537$, and their disjoint union has 358, 952 members.

The algorithms producing paths may be greedy. Proposition 16 requires only the validity and complete accounting of the recorded paths; it assumes nothing about discovery of every available reduction.

10.3 P: the polynomial computation and its lattice

For precision we specify the rational polytope used in P. Fix rank $n \geq 2$ and a positive-base triple T . Its hive has coordinates $h(a, b)$ for integers $a, b \geq 0$, $a + b \leq n$, with boundary

$$h(a, 0) = \sum_{j \leq a} \mu_j, \quad h(n - b, b) = |\mu| + \sum_{i \leq b} \nu_i, \quad h(0, b) = \sum_{j \leq b} \lambda_j.$$

Its three rhombus inequalities, whenever all displayed vertices exist, are

$$\begin{aligned} h(a+1, b) + h(a, b+1) - h(a, b) - h(a+1, b+1) &\geq 0, \\ h(a, b) + h(a+1, b) - h(a, b+1) - h(a+1, b-1) &\geq 0, \\ h(a, b) + h(a, b+1) - h(a+1, b) - h(a-1, b+1) &\geq 0. \end{aligned}$$

Eliminate the fixed boundary coordinates. The resulting polytope $H_T \subseteq \mathbb{R}^D$ uses the full lattice $\mathbb{Z}^D$, where $D = (n-1)(n-2)/2$, with free vertices ordered by increasing a , then b . Each inequality is stored as $(b_0, a_1, \dots, a_D)$, meaning $b_0 + \sum a_i z_i \geq 0$. This fixes signs, boundary placement and lattice.

Lemma 17 (Literal array–hive correspondence). *For every integer $t \geq 1$,*

$$\#(tH_T \cap \mathbb{Z}^D) = c_{t\mu, t\nu}^{t\lambda}.$$

Proof. We give the coordinate correspondence, consistent with Buch [6, Theorem 1 and Fulton’s appendix]. At stretch one put $M_j = \sum_{r \leq j} \mu_r$ and

$$F(b, j) = M_j + \sum_{r \leq j, k \leq b} x_{r, k}, \quad h(a, b) = F(b, a + b).$$

The array margins and triangular support give exactly the three boundaries above. Conversely extend $F(b, j) = h(j - \min(b, j), \min(b, j))$ and set

$$x_{r, k} = F(k, r) - F(k-1, r) - F(k, r-1) + F(k-1, r-1).$$

Telescoping gives the row and content sums and makes the maps inverse. For $s = a + b$, the three rhombus slacks in the displayed order become $C_{s+2, b+1}$, $L_{s+1, b}$, and $x_{s+1, b+1}$. Thus the nontrivial column and lattice inequalities and off-diagonal nonnegativity match. The omitted lattice cases vanish by triangular support; the remaining column cases are outer-part inequalities. Diagonal nonnegativity follows from $x_{n, n} = \nu_n \geq 0$ and $L_{r+1, r} = x_{r, r} - x_{r+1, r+1} \geq 0$. These arguments work over $\mathbb{R}$ and preserve integer coordinates in both directions. The array polytope is bounded by its nonnegative row sums, so H_T is bounded. All constants and maps are homogeneous in the boundary parts; replacing T by tT replaces H_T by tH_T for $t > 0$. Lemma 1 proves the count. $\square$

The computation of a full polynomial has two exact routes. The primary route is Normaliz 3.11.1 on these integer inhomogeneous inequalities, writing each row as $(a_1, \dots, a_D, b_0)$ for Normaliz, with the task `EhrhartSeries`; no sublattice or rescaled grading is supplied. The mathematical algorithm homogenizes the rational polytope, triangulates the resulting rational cone and counts full-lattice residue classes in simplicial cones. A disjoint half-open decomposition avoids counting shared faces twice. Its rational generating series can then be converted to the Ehrhart quasipolynomial. If a constituent has denominator $q > 0$, its displayed coefficient vector $(a_0, \dots, a_d)$ is read as $\sum_{i=0}^d (a_i/q) t^i$. For these hives polynomiality proves that every quasipolynomial constituent agrees with P_T : each agrees with P_T at infinitely many positive integers. The verifier requires a polynomial output, rejects a reported nontrivial period, and checks the degree and dimension bounds. The algorithm and its exact arithmetic implementation are documented in the Normaliz manual [12].

The independent route counts $tH_T \cap \mathbb{Z}^D$ with LattE for $t = 0, \dots, D+2$ and checks agreement with the reported rational polynomial. Since T has positive base count, H_T is nonempty and the count at zero is 1. Agreement at any $D+1$ of these arguments, together with degree at most D , identifies the polynomial; the additional arguments are checks. This route is complete for every

residual triple of rank at most five. At higher ranks, the existing dataset uses Normaliz as the full-polynomial engine; separate lrcalc checks at $t = 1$ and selected $t = 2, 3$ values test it but do not identify a polynomial of degree as large as fifteen.

Proposition 18 (Finite dependency P). *For each $T \in \mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{large}}$, the supplied record contains a rational coefficient vector $(a_{T,0}, \dots, a_{T,d_T})$, $d_T \leq 15$, which the specified exact Ehrhart computation identifies with P_T . Every entry is nonnegative. There is one accepted polynomial for every residual triple and no unresolved or mismatched source. Consequently*

$$\forall T \in \mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{large}}, \quad \forall t \in \mathbb{Z}_{\geq 1}, \quad c_{t\mu, t\nu}^{t\lambda} = \sum_{i=0}^{d_T} a_{T,i} t^i, \quad a_{T,i} \geq 0.$$

The polynomial-record audit checks source triples and residual membership, parses all fractions with positive denominators, evaluates all recorded values by rational arithmetic, recomputes all base counts, and tests every coefficient. It checks complete source coverage and rejects incompatible duplicate results. A historical failed computation remains visible and is superseded only by the separately identified successful result for the same triple and verifier source. Missing data, an interrupted call, a timeout or an engine error never establishes P.

The recorded polynomial audit has the following scope. The degree is the degree in the ordinary stretching variable.

	$\mathcal{C}_{\text{small}}$	$\mathcal{C}_{\text{large}}$
Residual members accounted for	12,415	346,537
Negative monomial coefficients	0	0
Unresolved residual computations	0	0
Degree range	2–10	1–13
Complete independent LattE reproduction	2,984	20,824
Normaliz as sole full-polynomial engine	9,431	325,713

The journals also contain nonresidual records; their presence does not inflate the residual cardinality. The small journal has 13,139 records in 13,383,091 bytes, and the large journal has 509,106 records in 544,824,662 bytes. The complete-prefix hashes and residual-index hashes below bind the data used in E, R and P. In each hash the two lines are concatenated without a space.

Artifact	SHA-256
Small residual indices	52b87cdec5f5f2fa61eba8af84484d5685 081625120730584b3c54f4d984b942
Large residual indices	8db1ff0e08c602b3d0cab8fdbcb5a1c5ef 33743fade2975e2a2337419cbcd31c0
Small polynomial journal	5f9fca8c6847a35cafb09d49499e99e13f aa4391018632288d2ea29513ffca67
Large polynomial journal	ec8fa3340c55a12da8ed8c4f822aa0129 030690e5ddd7d69c4f5b286bbb8a264
Large error-resolution record	1b7ec4392a6a05cac4ad6181c960f160d 64a09cb26008ce1a7335968161752a9

The last record resolves large source index 1,104,809, retaining the historical failure. It is included in the accepted residual count.

10.4 Complete evidence and fresh replay

The complete original evidence manifest binds 53 files, including the exact verifier, both full journals, every reduction path and both residual index sets. Its SHA-256 is the concatenation

ba391e5b641153601826a22056300533b
1c74fbb12eec0084150c888f789cff4.

A fresh seven-stage replay on 6 September 2026 verified all of those hashes, reproduced both complete domains and the local reduction cover, and checked all 358,952 residual polynomial records with freshly recomputed LR base counts. Every required comparison passed; elapsed time was 110.696 seconds. The replay receipt SHA-256 is the concatenation

dbda7ada1e15068bb5828da8ea0a37e3
71d233fb77e65c4d9e68bf379e98152a.

The accompanying bundle maps these records to E, R and P in EVIDENCE.md. From its extracted root, the command `python3 verify.py --output verification.json` checks the bundle manifest and repeats the complete finite audit. The separate resumable regeneration command in REPRODUCE.md reconstructs the exact hive inputs and reruns the Ehrhart algorithm, comparing its rational coefficients with each archived source record. Neither a subset regeneration nor the fast audit is described as a new exhaustive Ehrhart computation.

An independent literal-row check agreed with the source on the entire balanced-boundary vector space at each rank 1 through 7: both row builders are linear in the boundary, and they agree on a basis and zero. Actual backend controls checked a negative-coordinate interval, a lower-dimensional segment in the full ambient lattice, an empty polytope, two nontrivial quasiperiods, and a simplex with a negative linear Ehrhart coefficient. These controls test the interpretation and parser; they do not replace the complete residual computations.

Most original temporary Normaliz input/output files were removed by the producing program. The durable journals retain the source triples, exact coefficient vectors and checks; the inputs are reconstructed by the frozen source. The executable and linked-library census is a later measured snapshot, not an attestation captured before each historical call. These are exact-algorithm execution records with a reproducible regeneration route, rather than an independently checked cone certificate for every case.

10.5 Reproducibility and the trust boundary

The accompanying `epoch_stretched_lr_full_proof.zip` contains the complete domain lists, every coverage record, all residual indices and coefficient records, the exact source constructing the inequalities, the audit programs, the producing programs, and a manifest of file and executable identities. A replay receipt must specify which artifacts were rechecked and which computations were rerun. The manifest maps these artifacts to E, R and P and includes the historical-error resolution. File names are relative to the bundle; no author's local directory is part of the proof.

Replaying the finite audits establishes their explicit integer and rational assertions, conditional on correct execution of those programs. In particular, auditing saved coefficients and a few LR values does not replace the full Ehrhart calculation. At ranks six and seven that calculation relies on the pinned Normaliz implementation. The existing records do not contain independently checked cone decompositions for every residual member, and they are not a proof of Normaliz in a proof assistant. This is the explicit conventional computer-assisted trust boundary. Removing it requires a verified implementation or complete checked decomposition certificates; no such verification is asserted here.

Theorem 19 (Computed finite-box positivity). *For every balanced triple $T = (\lambda, \mu, \nu)$ of partitions satisfying*

$$\max\{\ell(\lambda), \ell(\mu), \ell(\nu)\} \leq 7, \quad \max\{|\lambda|, |\mu|, |\nu|\} \leq 30,$$

every coefficient of $P_{\mu\nu}^\lambda$ in the monomial basis is nonnegative. Thus no triple requested by the bounded problem exists.

Proof. The complete exact computation, with the stated software trust boundary, establishes E, R and P. E gives complete enumerations of the necessary domain. R gives a valid exclusion or residual membership for every enumerated triple. P gives nonnegative polynomials for all residual members. These are exactly the three premises of Theorem 14. $\square$

The supplementary mathematical appendix has readable-manifest SHA256

0c0cf5e476155d11228259ae515827179023f01e3aede851fd2bb6d35e663465

It additionally supplies a separate literal-condition enumeration, agreeing on all 1,292,758 domain triples and base counts after 1,587,401 fresh LR evaluations. Stratified reduction checks test 193 cases and all 579 identities at $t = 1, 2, 3$, including the final second-reduction exclusions and paths with out-of-box intermediate triples. These are supplementary execution checks; the all-stretch identities are proved above. The appendix contains the complete mathematical data and source programs and a restricted reader through which a referee can inspect them and run the finite audit.

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